Question	Answer	Marks	Partial Marks
17	3 230 or 3231 or 3232 or 3231.7…	**3**	**B2** for 9230 or 9231 or 9232 or 9231.7… OR **M2** for $6000\times\left(1+\frac{9}{100}\right)^5-6000$ oe or **M1** for $6000\times\left(1+\frac{9}{100}\right)^5$ oe
18(a)	47.8 or 47.77 to 47.78	**2**	**M1** for $\frac{172\times1000}{60\times60}$ oe or **B1** for figs 478 or 4777 to 4778
18(b)	2.33 or 2.325 to 2.326	**2**	**M1** for $\frac{176-172}{172}[\times100]$ oe or $\frac{176}{172}\times100[-100]$ oe or $\left(\frac{176}{172}-1\right)[\times100]$ oe
19(a)	3.47×10^{-8}	**1**	
19(b)	1.5×10^{11}	**1**	
20(a)	0	**1**	
20(b)	4.8 or $4\frac{4}{5}$	**2**	**M1** for $5y=3\times8$ or $\frac{y}{3}=\frac{8}{5}$ or better
21(a)	w^{30} final answer	**1**	
21(b)	t^9v^5 final answer	**2**	**B1** for t^9v^k or t^kv^5, final answer or correct answer seen then spoilt
22(a)	$8x-12$ final answer	**2**	**M1** for $2x-5+2x+2+4x-9$ soi or better or **B1** for $8x$ or -12, in final expression or correct expression seen then spoilt
22(b)	6.5	**2**	**M1** for $8x=40+12$ or $x-\frac{12}{8}=\frac{40}{8}$ or a correct first step in rearranging *their* $kx+j=40$ $k\neq0,\ j\neq0$

Question	Answer	Marks	Partial Marks
22(c)	60	**3**	**FT** *their* **(b)** for 3 marks if *their* **(b)** > 2.5 **M2** for $\frac{1}{2}\times(2\times \textit{their}\,\mathbf{(b)}-5)\times(2\times \textit{their}\,\mathbf{(b)}+2)$ oe or better or **M1** for substituting *their* **(b)** into $2x-5$ and $2x+2$ or **M1** for $\frac{1}{2}\times(2x-5)\times(2x+2)$
23	32	**4**	**M3** for [h =] $\dfrac{\frac{4}{3}\times\pi\times 8^3}{\frac{1}{3}\times\pi\times 8^2}$ oe or **M2** for $\frac{4}{3}\times\pi\times 8^3=\frac{1}{3}\times\pi\times 8^2\times h$ oe or **M1** for $\frac{4}{3}\times\pi\times 8^3$ oe or $\frac{1}{3}\times\pi\times 8^2\times h$ oe
24	12.8 or 12.76…	**5**	**M4** for tan 52 = $\dfrac{d}{\sqrt{14.7^2-10.8^2}}$ oe or **B3** for 9.97[…] or **M2** for $14.7^2-10.8^2$ oe or **M1** for $10.8^2+(BD)^2=14.7^2$ oe if M2 or M1 or M0 scored, **SC1** for $\tan 52=\dfrac{d}{\textit{their}\,BD}$ oe